

Coefficients as Partial Derivatives

Teaching note

Jennings (1990), *The Accounting Review*

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One label, incremental information content, covered two different hypotheses

“Therefore, the interpretation of results beyond a statistical statement that one variable has information content incremental to another depends on the specification of the regression equation that is estimated.”

Jennings (1990, p. 925)

By 1990 the phrase “incremental information content” covered a coefficient on cash flow given accruals and a coefficient on cash flow given income. Jennings’ objection is that “these studies have not described the accounting research implications of their findings beyond the statistical evidence of a conditional association with security returns” (p. 926). The two coefficients test different hypotheses, and the reason is a property of partial derivatives.

■ Incremental content conceals two hypotheses: composition and disclosure

Composition: which components belong in income. Under a returns criterion, “the most informative measure of income includes income components that are valued by investors” (p. 926). In a regression, a *zero restriction* on a component coefficient.

“For components that are valued by the market equivalently, disclosure of their sum is sufficient because investors are indifferent to which component contributed more or less to income.”

Jennings (1990, p. 926)

Disclosure: whether two components should be reported separately; an *equality restriction* between component coefficients.

Substituting the identity maps two specifications onto three unique restrictions

All variables are unexpected changes; intercepts suppressed.

$$\text{INC} = \text{CFO} + \text{TA} \quad (1) \quad \text{RET} = \theta_1 \text{CFO} + \theta_2 \text{TA} + \varepsilon \quad (2) \quad \text{RET} = \gamma_1 \text{CFO} + \gamma_2 \text{INC} + \varepsilon \quad (3)$$

Substitute (1) into (3):

$$\text{RET} = (\gamma_1 + \gamma_2) \text{CFO} + \gamma_2 \text{TA} + \varepsilon$$

$$\text{(T-1)} \quad \theta_1 = \gamma_1 + \gamma_2$$

composition of CFO

$$\text{(T-2)} \quad \theta_2 = \gamma_2$$

composition of TA

$$\text{(T-3)} \quad \theta_1 - \theta_2 = \gamma_1$$

disclosure of CFO versus TA

Because the regressors in (2) and (3) are linear transformations of each other, so are the coefficients (p. 927, with footnote 4 crediting Christie et al. 1984). Of the six coefficients across the three two-regressor specifications, “only three are unique” (p. 927).

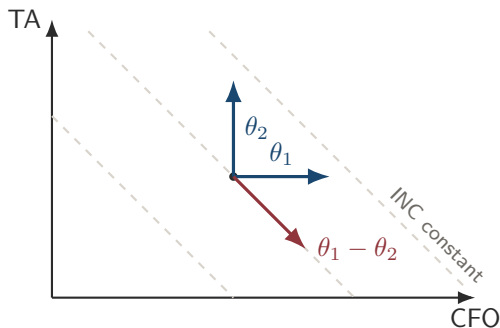
The conditioning variable fixes the direction in which the derivative is taken

A coefficient is $\partial \text{RET} / \partial x$ with the other regressor fixed; each conditioning choice is a direction.

TA fixed: horizontal; slope θ_1 .

CFO fixed: vertical; INC moves only through TA, since “a one-dollar change in unexpected income is identical to a one-dollar change in total accruals” (p. 928); slope $\gamma_2 = \theta_2$.

INC fixed: a dollar of TA is swapped for a dollar of CFO; slope $\gamma_1 = \theta_1 - \theta_2$.



When cash flow and accruals are valued equally, CFO given INC is zero by construction

```
set obs 1000
gen cfo = 0.15*rnormal()
gen ta = -0.4*cfo + 0.09*rnormal()
gen inc = cfo + ta
gen ret = 0.30*cfo + 0.30*ta ///
        + 0.20*rnormal()
reg ret cfo ta           // eq. (2)
test cfo = ta           // disclosure
reg ret cfo inc         // eq. (3)
test cfo                // the same F
```

	Economy 1: $\theta_1 = \theta_2 = 0.30$		Economy 2: $\theta_1 = 0.45, \theta_2 = 0.15$	
	Eq. (2)	Eq. (3)	Eq. (2)	Eq. (3)
CFO	0.34 (6.7)	-0.03 (-0.5)	0.49 (9.7)	0.27 (4.7)
TA	0.37 (5.5)		0.22 (3.2)	
INC		0.37 (5.5)		0.22 (3.2)
R^2	0.05	0.05	0.09	0.09

Coefficients (t -statistics); simulated data (seed 1990), $N = 1,000$, identical data in both specifications; code abridged from `scripts/jennings_reparam.do`. In Economy 1 CFO is informative ($t = 6.7$ in eq. 2) yet its eq. (3) coefficient is -0.03 ; $\gamma_1 = \theta_1 - \theta_2$, $\gamma_2 = \theta_2$, and the two F statistics agree to machine precision.

Which hypothesis a coefficient tests follows from its conditioning variable

“When TA is the conditioning variable (eq. [2]), CFO has incremental information content if it is an informative component of net income. When INC is the conditioning variable (eq. [3]), CFO has incremental information content if it is valued differently than TA, implying that investors prefer to observe CFO and TA disclosed separately.”

Jennings (1990, p. 928)

Coefficient	Conditioning variable (held fixed)	Restriction	Hypothesis
CFO in (2)	TA	$\theta_1 = 0$	composition of CFO
TA in (2)	CFO	$\theta_2 = 0$	composition of TA
CFO in (3)	INC	$\gamma_1 = \theta_1 - \theta_2 = 0$	disclosure of CFO versus TA
INC in (3)	CFO	$\gamma_2 = \theta_2 = 0$	composition of TA

Aggregate-plus-component specifications require translation before they can be read

Bowen et al. (1987) regress returns on four overlapping measures linked by three identities, $INC = CFO + CA + NCA$, $WCFO = CFO + CA$, and $CFAI = CFO + INV$:

$$RET = \beta_1 INC + \beta_2 WCFO + \beta_3 CFO + \beta_4 CFAI + \varepsilon \quad (9)$$

Jennings' component form (10), $RET = \gamma_1 CFO + \gamma_2 CA + \gamma_3 NCA + \gamma_4 INV + \varepsilon$, has $\gamma_1 = \beta_1 + \beta_2 + \beta_3 + \beta_4$, $\gamma_2 = \beta_1 + \beta_2$, $\gamma_3 = \beta_1$, and $\gamma_4 = \beta_4$; a joint test on two β 's restricts a different set of γ 's.

"The interpretation of these results in terms of the composition and disclosure of these components of income (CFO, CA, NCA, and INV) is not obvious from this specification."

Jennings (1990, p. 930)

The marginal effect of interest is a chain-rule combination of coefficients

For $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, the coefficient β_1 is $\partial y / \partial x_1$ with x_2 held fixed. If the accounting question moves x_2 with x_1 along $x_2 = g(x_1)$, the relevant derivative is

$$\frac{dy}{dx_1} = \beta_1 + \beta_2 \frac{dg}{dx_1}$$

Setting	Regression	Reported coefficient holds fixed	Marginal effect in the alternative direction
Jennings eq. (3)	RET on CFO, INC	γ_1 : INC (TA falls one for one)	$\theta_1 = \gamma_1 + \gamma_2$ (TA fixed; composition of CFO)
Interaction term	RET on E , Size, $E \times \text{Size}$ ($x_2 = E \times \text{Size}$)	β_1 : Size = 0	$\beta_1 + \beta_3$ Size at a stated Size
Ohlson levels	MV on E , BV_t	α_1 : BV_t (a dollar of E paid out)	$\alpha_1 + \alpha_2$ (a dollar of E retained)
BGL Panel (A)	R on HC, Z	b_1 : Z (PRE moves c per dollar of HC)	$b_1 = \beta_1 - c\beta_2$ (PRE fixed)

In every row the test of the accounting hypothesis is a test of a linear combination of coefficients, evaluated at a stated point and in a stated direction (`lincom, test`).

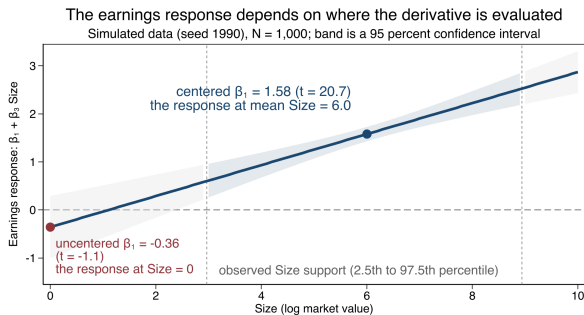
In an interaction model, β_1 is the earnings response at $\text{Size} = 0$

$$\text{RET} = \beta_0 + \beta_1 E + \beta_2 \text{Size} + \beta_3 E \times \text{Size} + \varepsilon$$

$$\partial \text{RET} / \partial E = \beta_1 + \beta_3 \text{Size}$$

$$\partial \text{RET} / \partial \text{Size} = \beta_2 + \beta_3 E$$

Mean centering, $\text{Size} - \overline{\text{Size}}$ and $E - \overline{E}$, leaves β_3 unchanged and makes β_1 the response at the sample mean. Here the uncentered β_1 is -0.36 ($t = -1.1$) and the centered β_1 is 1.58 ($t = 20.7$): the same line read at two points, one of them outside the data.



Under clean surplus, a dollar of earnings moves value by

$$\alpha_1 + \alpha_2$$

$$MV = \alpha_0 + \alpha_1 E + \alpha_2 BV_t + \varepsilon$$

$$BV_t = BV_{t-1} + E - Div \quad (\text{clean surplus, A2a})$$

$$\partial MV / \partial E \big|_{BV_{t-1}, Div} = \alpha_1 + \alpha_2$$

α_1 holds BV_t fixed: a dollar of earnings matched by a dollar of dividends.

Regressing MV on E and $K = BV_{t-1} - Div$ is an exact reparameterization and returns $\alpha_1 + \alpha_2$ on E , with the same residuals and R^2 .

	(1) <i>MV</i> on <i>E</i> , <i>BV_t</i>	(2) <i>MV</i> on <i>E</i> , <i>K</i>
<i>E</i>	4.97 (73.6)	5.77 (95.3)
<i>BV_t</i>	0.80 (40.7)	
<i>K</i> = <i>BV_{t-1}</i> - <i>Div</i>		0.80 (40.7)
<i>R</i> ²	0.93	0.93

Simulated data (seed 1995), $N = 1,000$, generated as $MV = 2 + 5.0 E + 0.8 BV_t + e$ with BV_t built from clean surplus. Column (2) rewrites column (1); the coefficient on E is $\alpha_1 + \alpha_2$ exactly, with the `lincom` t -statistic.

In BGL's Panel (A), the coefficient on HC is the derivative along the first-stage line

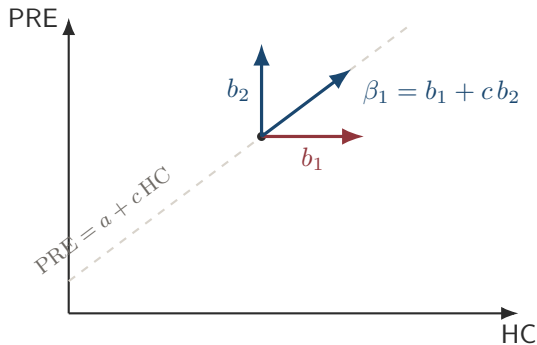
Stage 1: $PRE = a + c HC + Z$

Stage 2: $R = \alpha + \beta_1 HC + \beta_2 Z + u$

With $R = b_0 + b_1 HC + b_2 PRE + e$ and
 $PRE = a + c HC + Z$:

$$\beta_1 = b_1 + c b_2, \quad \beta_2 = b_2$$

Holding Z fixed and raising HC by a dollar raises PRE by c : β_1 is the derivative along the fitted line. With BGL's Table 4 numbers,
 $0.39 + 1.46 \times (-0.02) = 0.36$.



One regression yields three derivatives with respect to HC, one per variable held fixed

Held fixed	Derivative	Estimate (t)	Where it appears
PRE	b_1	0.42 (4.3)	Panel (B) residual; one regression
$D = \text{PRE} - \text{HC}$	$b_1 + b_2$	0.39 (6.5)	HC in R on HC, D
$Z = \text{PRE} - a - c\text{HC}$	$b_1 + c b_2, c = 1.50$	0.38 (7.1)	Panel (A) HC; bivariate slope

```
reg r hc pre
lincom hc           // PRE fixed
lincom hc + pre     // D fixed
lincom hc + 1.50*pre // Z fixed
```

Simulated data, $N = 303$, the seed-228 draw of the BGL module ($c = 1.50$; BGL's 1977 estimate is 1.46); D is the depreciation adjustment, $\text{PRE} = \text{HC} + D$.

Mapped onto Jennings' (CFO, TA, INC): b_2 (PRE given HC) is the composition test for D , and b_1 (HC given PRE) equals the HC coefficient minus the D coefficient in R on HC and D , the disclosure test of HC against D .

Relabeled, the tests support composition but not disclosure

Jennings' re-tests of Rayburn (1986) and Bowen et al. (1987) yield “consistent and strong evidence that both cash flow and accrual components add to the informativeness of income” (p. 931); nine pairwise equality tests reject nothing at 5 percent.

“However, there is only weak and inconsistent evidence that accrual components of income are valued differently from cash flow components by investors.”

Jennings (1990, p. 931)

A skeptic would note that equality of long-window slopes is a narrow criterion (linear, single period) and that 20 or 10 annual estimates carry little power; the null may reflect power rather than indifference.

Summary

- A regression coefficient is a partial derivative taken in a direction fixed by the other regressors. The label “incremental information content” therefore describes a specification, not an accounting hypothesis; the hypothesis follows from what the conditioning variable holds fixed.
- When regressors are linked by an identity, a clean-surplus relation, a first-stage line, or an interaction, the marginal effect that answers the accounting question is a linear combination of coefficients evaluated at a stated point, and the hypothesis test belongs on that combination rather than on a single coefficient.
- Applied to Rayburn (1986) and Bowen et al. (1987), the relabeling yields strong evidence that cash flow and accrual components each add to the informativeness of income and weak, inconsistent evidence that investors value them differently.

Jennings (1990) · Christie et al. (1984) · Beaver, Griffin, and Landsman (1982) · Ohlson (1995) · github.com/sbb151/acctg597a-research-design